

Predictive and feedback signals differently shape the formation of language representations

Shuguang Yang ^{1,2}, Shaoyun Yu ^{3,4}, Xin Jiang ^{1,2}, Suiping Wang ^{1,2}, Gangyi Feng ^{3,4*}

1. School of Psychology, South China Normal University, Guangzhou 510631, China
2. Philosophy and Social Science Laboratory of Reading and Development in Children and Adolescents (South China Normal University), Ministry of Education, Guangzhou 510631, China
3. Department of Linguistics and Modern Languages, The Chinese University of Hong Kong, Hong Kong SAR, China
4. Brain and Mind Institute, The Chinese University of Hong Kong, Hong Kong SAR, China

* Corresponding author

Gangyi Feng

Brain and Mind Institute

Department of Linguistics and Modern Languages

The Chinese University of Hong Kong, Hong Kong SAR, China.

Email: g.feng@cuhk.edu.hk

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Abstract

Adults vary greatly in how effectively they learn a new language, but the signals driving the learning processes and individual differences remain unclear. Over seven days, we tracked behavioral learning and collected fMRI data from 102 adults as they learned an artificial language with corrective feedback. We trained matched transformer models with prediction, feedback, or combined objectives and compared their internal representations to brain activity. Representations derived from the prediction-focused model accounted for the largest share of unique neural variance at the group level, despite the human task being feedback-based. Throughout model training, both objectives showed a shift in brain-model alignment from sensory to higher-order language and associative networks, indicating abstraction processing. Conversely, neural information related to feedback was most useful for predicting individual generalization outcomes on Day 7. These findings support a multi-signal model of adult language learning, in which prediction shapes a common neural learning architecture across learners, whereas feedback-related mechanisms better explain individual differences over time.

Keywords: language learning, individual difference, prediction, feedback signal, neural systems

Introduction

Adults often spend years learning a new language, yet progress is often slow, effortful, and varies greatly among individuals. Even with similar instruction and exposure, some learners rapidly extract linguistic regularities and generalize them to new materials, whereas others improve more slowly or reach only modest levels of proficiency (Kidd et al., 2018; Krishnan et al., 2016; Morgan-Short, 2020). Understanding the language learning process and its variability is key to understanding human learning, as it affects not only language acquisition but also how the adult brain builds new, structured knowledge from experience.

A longstanding debate centers on whether adult language learning is driven primarily by a single learning process or by multiple signals that contribute differently depending on tasks. Two candidates have been especially prominent. One highlights prediction-based learning, where learners track distributional structure in the input and form expectations about upcoming sounds, words, and syntactic relationships. This enables them to build representations that support generalization across different contexts and time frames (Caucheteux et al., 2023; Heilbron et al., 2022; Pelucchi et al., 2009; Saffran, 2002; Saffran et al., 1996). The other underscores feedback-based learning, in which explicit correctness signals and outcome-dependent adjustments help learners refine responses and acquire abstract constraints. This account emphasizes the role of cortico-striatal systems involved in sequential and category learning (Feng, Gan, et al., 2021; Feng, Ou, et al., 2021; Krishnan et al., 2016; Schultz et al., 1997), considering the fundamental process of language learning. Both perspectives are plausible and supported by empirical evidence. However, it remains unclear how these learning signals shape the formation of language representations and whether they matter differently for the formation of common neural organization versus individual differences among learners.

This uncertainty persists in part because the two accounts are usually studied separately. Prediction-based learning is often examined using statistical learning or self-supervised language modeling (Bovolenta & Marsden, 2022; Ito et al., 2018; Rabagliati et al., 2015), whereas feedback-based learning is typically studied using supervised or reinforcement-like paradigms (Arbel et al., 2014, 2017). These different approaches use different tasks, models, and outcome measures, making direct comparison challenging. As a result, three questions remain unresolved. First, does adult language learning rely on multiple learning signals instead of a single dominant mechanism? Second, do these signals contribute differently across brain systems and across stages of learning? Third, can these differences explain why some learners generalize successfully while others do not?

Here, we address these questions using a framework for comparing candidate learning signals in a shared representational space. The central innovation is to combine matched-architecture transformer models, representational similarity analysis, and individualized prediction within the same learning paradigm. This design allows us to manipulate learning objectives while keeping the model architecture and training

materials constant, enabling us to measure how the resulting internal representations correspond to neural activity during human learning. It also allows us to test whether mechanism-related neural information can better predict individual learning outcomes. Therefore, this study examined not only whether a model aligns with the brain but also whether this alignment can distinguish among candidate mechanistic theories of human language learning.

A key conceptual point is that the current models focus on learning signals rather than fully implementing entire theoretical frameworks. Specifically, we operationalize the prediction objective as next-token prediction and the feedback objective as correctness-supervised grammaticality judgment using a trial-wise feedback structure. These objectives capture distinct pressures on the learner. The prediction objective imposes an internally generated pressure to anticipate upcoming structure from distributional regularities, whereas feedback imposes an externally provided pressure to update responses based on success and error signals. Therefore, they should not be viewed as direct, one-to-one implementations of predictive coding theory (Friston, 2005) or algorithmic reinforcement learning (Sutton & Barto, 2018) in their strongest forms. Instead, they serve as controlled approaches to test whether representations optimized for prediction and feedback objectives relate differently to human neural activity and behavior. This distinction is especially important because prediction can either be a driver of language learning or an emergent byproduct of it (Chang et al., 2006, 2013). A comparative framework is necessary to determine which interpretation is better supported by model-brain alignment during learning.

Transformer models are particularly useful for this purpose. Classical symbolic, feature-engineered and statistical models have provided important mechanistic insights, but they typically rely on hand-crafted representations or modular pipelines that are less suited to capturing end-to-end representational change during learning (Elman, 1990; Naselaris et al., 2011; Riesenhuber & Poggio, 1999). By contrast, transformer models can be trained with different objectives on the same corpus, yielding directly comparable internal representations within a shared architecture. Recent work has also shown that such representations align with human neural activity across the cortical hierarchy during language processing (Caucheteux et al., 2022, 2023; Goldstein et al., 2022). This makes them a powerful tool for testing where, when and to what extent distinct learning objectives resemble the representational geometry of the human brain.

We applied this framework to a seven-day artificial-language learning study using Brocanto2 (Morgan-Short et al., 2010, 2012). We tracked behavioral learning in 102 participants over the full training period and collected fMRI data on Day 1, while participants performed grammaticality judgments with trial-by-trial corrective feedback. In parallel, we trained three GPT-style models with the same architecture but different objectives (Fig. 1). One was a prediction model optimized for next-token prediction, another was a feedback model optimized for grammaticality classification with corrective feedback, and the third was a combined model optimized for both tasks. We then aligned model representations with fMRI data using representational similarity

analysis (RSA) and variance partitioning, tracked stage-specific representational updates across training, and built individualized multimodal models that combined early neural and behavioral information to predict generalization performance on Day 7.

This design enabled us to test three specific predictions. First, if prediction-based learning provides a dominant representational scaffold during learning, then representations optimized for a prediction objective should explain a larger share of unique neural variance at the group level than those optimized for a feedback task, despite the human task being feedback-based. Second, if learning-related representations become more abstract during training, then model-brain alignment should move from lower-level sensory systems to higher-order language and associative networks. Meanwhile, these shifting patterns may vary between the two model types. Third, if learner-specific outcomes depend more heavily on how learning signals are encoded and used, then neural information related to a given learning signal should better predict later individual differences in generalization.

By jointly analyzing shared neural and model structures, as well as individual differences, this study aims to move the debate beyond the question of whether prediction- and feedback-based accounts are individually plausible. Instead, it investigates whether they contribute differently to human language learning and if those contributions arise at different levels of learning processing. More broadly, the work uses model-brain alignment not just as a descriptive comparison between artificial and biological systems but as a strategy for testing mechanistic hypotheses about how adults acquire new structured knowledge and why learners vary so greatly in their outcomes.

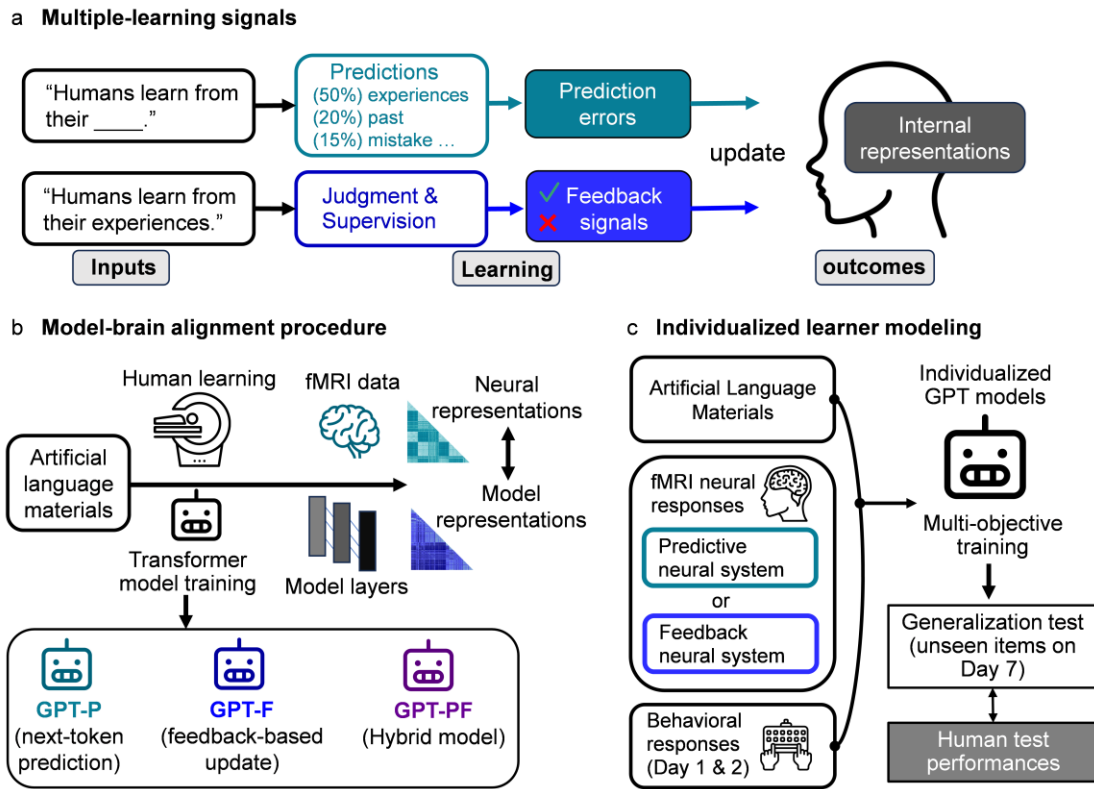


Fig. 1. Overview of the study framework and procedure. **a**, Hypothesized learning mechanisms supporting human language learning. The framework compares two candidate signals: prediction, based on next-element prediction among alternatives, and feedback-based learning, based on outcome signals (correct or incorrect). Both influence the internal representations for language learning. **b**, Model-brain alignment procedure. Participants learned an artificial language while fMRI data were collected, and parallel GPT models were trained on the language materials. Three models were compared: GPT-P (next-token prediction), GPT-F (feedback-based learning), and GPT-PF (both). RSA compared model layers' representations with neural responses. **c**, Individualized learner modeling. To explain individual differences in learning, participant-specific GPT models combined artificial-language materials with early behavioral responses and fMRI responses from prediction or feedback neural systems. These models predicted Day 7 generalization performance with unseen items. This approach examines which learning mechanism and neural system best explain the variation in language-learning success.

Results

Human and model language-learning performance

We collected fMRI data on Day 1 from 102 adults performing an artificial language-learning task that required grammaticality judgments with trial-by-trial correctness feedback. Participants then completed six additional days of behavioral training using the same paradigm, followed by a Day 7 generalization test that assessed

grammaticality knowledge on unseen items (Fig. 2a).

To test the candidate learning mechanisms and their possible integration, we trained three small GPT-style models with identical architectures and hyperparameters, differing only in training objective. GPT-P was trained for next-token prediction using cross-entropy loss. GPT-F was trained on the grammaticality-judgment task with explicit correctness feedback, allowing feedback-dependent updating without committing to a specific reinforcement-learning algorithm. GPT-PF combined both objectives by jointly optimizing next-token prediction and feedback-based judgment with dynamic loss weighting (see Methods). This design allowed us to isolate the contribution of each mechanism and test whether combining them produced better performances, while holding the model architecture and training data mostly constant.

All three models reached a performance plateau within 200 training epochs. GPT-P achieved 91% accuracy in generating unseen grammatical sentences, GPT-F reached 100% accuracy on grammaticality judgments, and GPT-PF achieved a balanced outcome of two measures, 96% generation accuracy and 97% grammaticality-judgment accuracy (Fig. 2c). The latter two models outperformed the peak Day 7 grammaticality-judgment accuracy of human learners (79%; Fig. 2b and Supplementary Fig. S1). GPT-PF outperformed the single-objective GPT-P model in sentence generation while matching GPT-F in grammaticality judgment, indicating that prediction and feedback-based learning make complementary contributions.

To examine how model behavior emerged, we tested whether changes in internal representations tracked changes in behavior across training checkpoints (see Methods and Fig. S2). Representational-update RDMs were constructed and positively correlated with behavior-update RDMs in all three models (Fig. 2d). In GPT-P, this coupling was significant across layers, but weaker than in GPT-F. In GPT-F, the association was robustly positive and strongest in the two hidden layers, indicating tighter coupling between internal updating and behavioral improvement. In GPT-PF, representational updates were positively related to both behavioral measures, with effects increasing across layers. Thus, learning in all three models involved systematic internal reorganization, but this reorganization was more tightly linked to model behaviors in GPT-F than in GPT-P.

We next compared representational geometry across models (Fig. 2e). GPT-P and GPT-F showed minimal similarity in both layers ($r_{\text{layer1}} = 0.081$, $r_{\text{layer2}} = 0.076$), indicating that prediction- and feedback-based objectives produced distinct internal representations despite similar training exposure. GPT-PF shared structure with both single-objective models, but its layer representations were more similar to GPT-F ($r_{\text{layer1}} = 0.357$, $r_{\text{layer2}} = 0.543$) than to GPT-P ($r_{\text{layer1}} = 0.298$, $r_{\text{layer2}} = 0.272$), especially in Layer 2. Thus, the combined model retained a stronger feedback-related representational signature while still incorporating prediction-based structure.

We then examined to what extent GPT-P and GPT-F aligned with human learning behavior across model training (Fig. 2f). In Layer 1, GPT-P showed stronger model-behavior alignment at the early stages (i.e., the first two checkpoints: $p = 2.82 \times 10^{-3}$ and 4.56×10^{-5}), did not differ reliably from GPT-F at the third checkpoint ($p = 0.34$),

and was weaker than GPT-F at the final checkpoint ($p = 1.99 \times 10^{-21}$). In Layer 2, GPT-P remained more strongly aligned through the first three checkpoints ($p = 2.60 \times 10^{-6}$, 1.80×10^{-6} and 6.60×10^{-3}), whereas GPT-F showed stronger alignment at the final checkpoint ($p = 3.11 \times 10^{-15}$). These results reveal a temporal shift in model-behavior alignment, in which prediction-based representations better captured early-to-intermediate learning, whereas feedback-based representations better captured later behavior as model performance approached the asymptote.

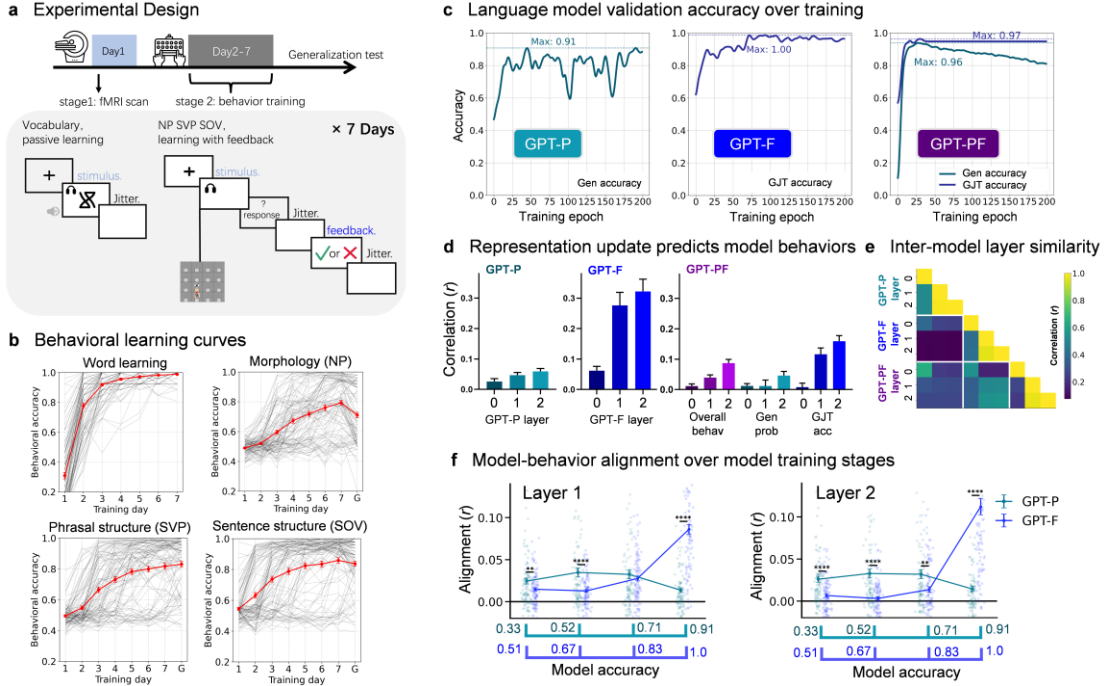


Fig. 2. Experimental design, behavioral learning curves, and model-behavior alignment. **a**, Seven-day training and neuroimaging procedure. Participants underwent fMRI on Day 1 during vocabulary and grammar training, followed by behavioral training on Days 2-7, ending with a generalization test. **b**, Behavioral learning curves over the 7 training days for word, morphology, phrasal, and sentence structure. Gray lines represent individual participants; red lines show the group means. G = generalization scores. **c**, Validation performance of the three models across training epochs. For GPT-P and GPT-PF, performance is displayed as the rate of generating novel grammatical sentences (“Gen accuracy”); for GPT-F and GPT-PF, grammaticality judgment test (GJT) accuracy is shown where applicable. **d**, Associations between model representational and behavioral change are shown with bar plots of Spearman correlations between representational-update RDMs and behavior-update RDMs across layers and models. **e**, Similarity of internal representations across models. The heat map displays layer-wise correlations between model RDMs. **f**, Model-to-human-behavior representation alignment. Mean RSA alignment (r) between behavioral patterns and model representations is shown across four model stages. Dots indicate individual participants, and error bars indicate SEM. Asterisks indicate paired t-test results for the

comparison between GPT-P and GPT-F at each stage. **, $p < 0.01$; ****, $p < 0.0001$.

Predictive representations accounted for most neural variance during learning

We next examined how closely the representational geometry of the three models aligned with human neural RDMs during learning. To do this, we used whole-brain searchlight RSA (Kriegeskorte, 2008; Stolier & Freeman, 2016) implemented with multiple regression (Fig. 3a). Layer-wise RDMs from the models were fitted to fMRI-derived RDMs while controlling for three nuisance RDMs (i.e., an untrained GPT model, stimulus length, and block identity). To characterize the relative contributions of the candidate mechanisms, we complemented whole-brain RSA with variance partitioning (Groen et al., 2018) and network-wise RSA across 12 canonical functional brain networks (Ji et al., 2019).

All three models showed reliable alignment with neural representations, with broadly similar spatial patterns across layers within each model (Fig. 3b and Fig. S3). However, the extent and distribution of this alignment differed across models. Although the human task focused on grammaticality judgments with feedback, the GPT-P model accounted for the greatest amount of neural variance (Fig. 3c). We also matched the number of training items used by GPT-F by additionally inputting the minimally corrected ungrammatical item and found consistent and robust model-brain alignment (Fig. S4). These alignments were observed not only in cortical regions but also in subcortical structures, including the caudate, nucleus accumbens and globus pallidus, consistent with a contribution of the basal ganglia to language learning (Krishnan et al., 2016). By contrast, the feedback model (GPT-F) aligned more closely with canonical perisylvian language regions (Hagoort, 2019) and the default mode network (Menon, 2023), whereas the integrated model (GPT-PF) showed additional alignment only in early sensorimotor cortices.

Variance partitioning further clarified these differences. Although participants performed an explicit grammaticality-judgment task, GPT-P accounted for the largest share of unique neural variance across model layers (Fig. 3c). We then tested whether this pattern generalized across large-scale systems by computing partial Pearson correlations between model RDMs and neural RDMs within the 12 canonical brain networks (Ji et al., 2019). Consistent with the overall variance-partitioning results, GPT-P showed significantly higher alignment than the other models in most networks (8 of 12; paired t -tests; Fig. 3d). These convergent analyses indicate that prediction-based representations capture the dominant group-level structure of neural representations during language learning.

To explore if GPT-F's neural alignment stems from reinforcement-like reward learning, we trained a pure policy-gradient reinforcement learning model (i.e., GPT-R). This was crucial because a model might perform well by maximizing reward but still have different internal representations from human learners. We examined whether GPT-F's brain alignment persisted when learning was reduced to reward-based optimization rather than instructional feedback. We replaced GPT-F and GPT-PF's

layer-wise RDMs with those of their RL counterparts (GPT-R and GPT-PR) within the same regression framework, while keeping the other regressors constant. While GPT-R achieved similar model accuracy, its internal representations hardly matched human neural activity (Fig. 3b, dashed box; see Supplementary Fig. S5 for more details), with no significant brain alignment across all voxels ($p > 0.05$). These results suggest that reward-based training alone was insufficient to reproduce the human neural representational geometry observed during language learning.

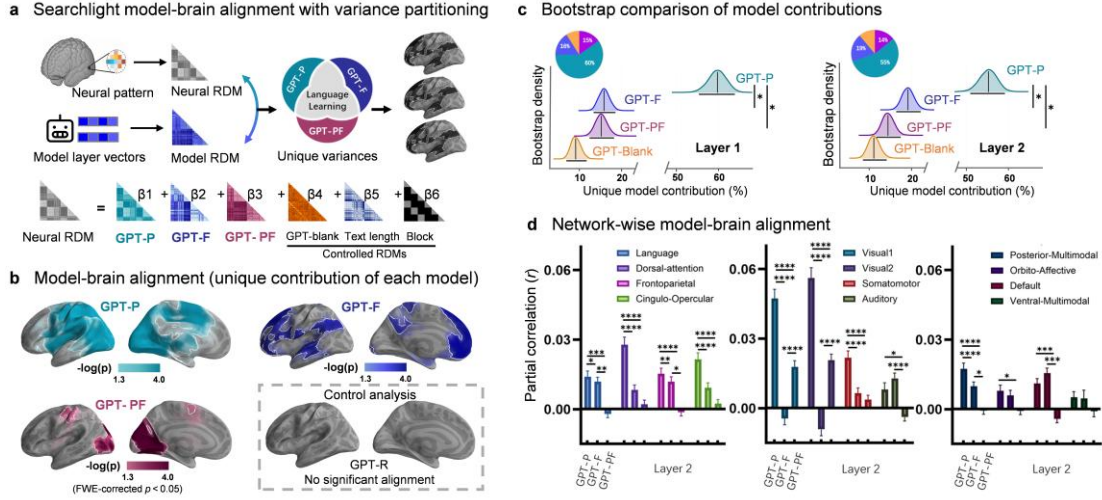


Fig. 3. Whole-brain and network-wise RSA dissociate the contributions of prediction and feedback-based learning. **a**, Schematic overview of the whole-brain searchlight RSA and variance-partitioning framework. At each searchlight location, neural RDMs were regressed against model RDMs to estimate the contributions of the models to neural representations. Neural geometry was modeled using three model RDMs (GPT-P, GPT-F, GPT-PF), controlling for an untrained GPT baseline (GPT-Blank), text length, and experimental block. **b**, Whole-brain searchlight RSA results for models' layer 2. In the control analysis (dashed box), the policy-gradient reinforcement model (i.e., GPT-R) showed no significant alignment, despite robust model performance (see Fig. S5). Brain maps were thresholded using TFCE with 10,000 permutations and family-wise-error corrected $p < 0.05$. **c**, Bootstrap comparison of relative RDM contributions. Bootstrap distributions of the proportion of total explained variance attributable to each model RDM across the first and second model layers. Each bootstrap sample's R^2 for each model RDM was shown as a percentage of the total R^2 from GPT-P, GPT-F, GPT-PF, and GPT-Blank. Density plots show resampled contribution estimates, with horizontal lines for central tendency and asterisks indicating significant differences. **d**, Network-level RSA across 12 canonical functional networks for the models' layer 2. Bars show partial correlations between neural and model RDMs within each network, controlling for the other RDMs. Prediction-based representations showed the strongest alignment across sensorimotor, language, and association networks, whereas feedback-based representations contributed more selectively. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

Overall, these findings support the multiple learning signal account during language learning and further challenge views that prediction is unnecessary for language learning (Huettig & Mani, 2016) or only a secondary effect of learning (Chang et al., 2013). Instead, prediction appears to be a central, relatively automatic force driving representational organization during adult language learning, even in feedback-based tasks. Feedback-based learning still provides additional information, but at the group neural level, the main signal remains predictive.

Human-like abstract model representations emerge progressively across training along a bottom-up cortical gradient

To determine when and how model representations become more human-like during model training, we performed RSA on representation updates rather than on static representations alone. Because human fMRI data were available only on Day 1, the neural RDMs provided a fixed reference for the brain’s representational geometry during early acquisition, rather than a temporally matched series across learning. We therefore used successive model checkpoints to ask at what point in model training the model’s internal changes began to resemble this early human learning state.

We sampled four checkpoints at equal intervals of improvement in validation accuracy, thereby dividing training into three stages (Fig. 4a and see Methods for details). For each stage, we computed representational change vectors and derived RDMs summarizing the structure of these updates. We focused on update RDMs rather than checkpoint RDMs because our question concerned the emergence of human-like representations during learning, not simply their presence at particular stages. Checkpoint RDMs capture the cumulative representational state of the model and conflate newly acquired structure with stable geometry inherited from earlier training. By contrast, update RDMs isolate the structure of representational change between adjacent stages, allowing us to test whether the learning process itself progressively becomes more similar to the organization of human neural representations. This made it possible to ask when and how models acquire human-like representational geometry, even though the human neural data were limited to the early stage of learning.

This analysis revealed that human-like model representations did not emerge all at once. Instead, they developed progressively during training, following a bottom-up cortical gradient, moving from lower-level sensory systems toward higher-order language and associative systems. For GPT-P, alignment between model update and neural RDMs gradually shifted from primary cortical regions to the language and default mode networks across training (Fig. 4b, lower panel), indicating that the model first captured relatively low-level representational structure and only later acquired representational structures resembling those in higher-order systems. To quantify this trajectory, we assessed partial Pearson correlations between model RDMs and neural RDMs within predefined brain networks (Ji et al., 2019). In GPT-P (Fig. 4c), update-based alignment decreased across stages in lower-order regions such as the secondary visual network, while increasing in higher-order systems such as the language and

associative networks (Fig. 4d-e). These findings indicate that the prediction-based model develops human-like representational geometry through gradual abstraction, with initial updates aligning with sensory systems and later updates resembling higher-level language and associative networks.

GPT-F showed a related but weaker and delayed pattern. Similar to GPT-P, it showed early alignment in lower-order systems, with significant effects first appearing in the secondary visual network and the adjacent ventral multimodal network starting at stage 2. Over time, this alignment also extended toward higher-order cortical systems. However, GPT-F showed no significant alignment in any network at stage 1, and overall, its brain alignment was lower than that of GPT-P (Fig. 4b, lower panel; Fig. 4c for network-level RSA results). Thus, both models showed a sensory-to-associative progression, but the emergence of human-like representational structure was earlier, stronger, and more continuous in GPT-P than in GPT-F.

To further quantify these temporal dynamics, we fitted a linear regression model to each participant's partial alignment values across the three training stages, separately for each model and network. This yielded a per-subject slope indexing the rate at which model-brain alignment increased or decreased over training. Comparisons of these slopes showed that, in both the language and default mode networks, GPT-P exhibited significantly steeper positive slopes than GPT-F (paired Wilcoxon tests, $p_{\text{Language}} = 2.40 \times 10^{-7}$, $p_{\text{DMN}} = 3.10 \times 10^{-10}$; Fig. 4e). This result confirms that the prediction-based model not only reached stronger alignment overall, but also more rapidly acquired update patterns resembling those of higher-order human brain systems.

These findings clarify when and how language models become human-like. Human-like representations develop gradually during training through a bottom-up reorganization from sensorimotor to higher-order systems. Both prediction- and feedback-based learning follow this hierarchy, though differently. GPT-P aligns well across all stages, indicating that its predictions support the emergence of human-like representations. GPT-F shows weaker, later-emerging alignment, suggesting that feedback-based learning becomes more human-like after sufficient training.

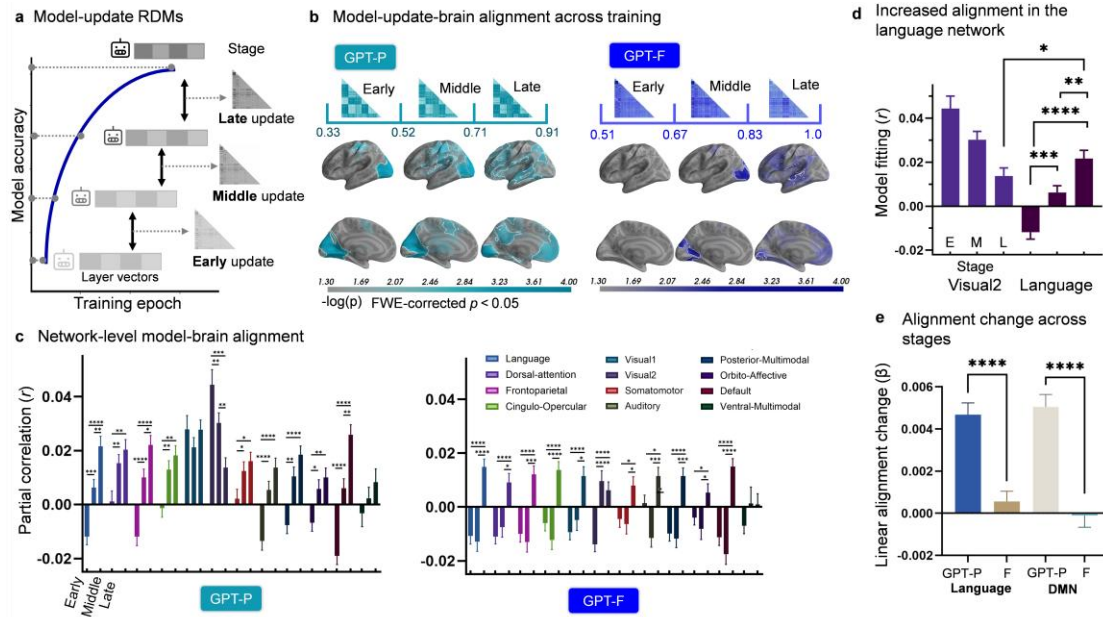


Fig. 4. Model-brain representation alignment shift from sensorimotor to language and higher-order systems. **a**, Schematic showing representation-update-based brain-model alignment analysis. For each adjacent checkpoint pair, item-wise hidden-state change vectors were computed, and representational-update RDMs were constructed from distances among these change vectors. This captures how internal model geometry updates across training stages. **b**, Whole-brain RSA results for representational updates in the first layer of GPT-P and GPT-F across three successive training stages. The numbers below the brain maps show model performance for sampled checkpoints. Brain maps are thresholded using TFCE with 10,000 permutations and FWE-corrected $p < 0.05$. **c**, Network-wise RSA of representational updates across training stages for both models. Bars show partial correlations between neural RDMs and stage-specific update RDMs within each of the 12 canonical functional networks. **d**, Direct comparison of GPT-P update alignment in the secondary visual (Visual2) and language networks. **e**, Linear slopes fitted to each participant's stage-wise update alignment values. GPT-P showed significantly larger increases than GPT-F in both the language and default mode networks, indicating a stronger late-emerging shift towards higher-order representational systems. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

Personalized neurocomputational models identify the drivers of individual differences

We further asked which mechanism better explains individual differences in the final learning outcomes. To address this question, we developed personalized GPT models, each trained on a participant's data using multimodal information from the early phase of human learning (Fig. 5a). The aim was to test whether different types of learner-specific neural information carry distinct explanatory value for later generalization. Each personalized model integrated three sources of information: early behavioral

responses, low-dimensional neural features extracted from selected regions (GPT-F- or GPT-P-related), and next-token predictions. This framework allowed us to evaluate which neural constraints were most informative for explaining why some learners generalized better than others.

These individualized models captured trial-level behavioral responses and, when trained with neural objectives, showed greater similarity to participant-specific neural responses than the behavioral-only model (Fig. 5b, c). We then asked whether they could predict individual generalization to unseen materials on Day 7. Early behavior alone provided limited predictive power ($r = 0.27$). By contrast, individualized models incorporating neural constraints improved prediction, reaching a cross-validated Spearman correlation of $r = 0.43$ ($p = 6.86 \times 10^{-6}$; Fig. 5d). In bootstrap comparisons, the model trained with feedback-related neural regions outperformed the behavioral-only model ($p = 0.0009$), as did the model trained with the combined neural mask ($p = 0.024$), whereas the model trained with prediction-related neural regions did not ($p = 0.17$; Fig. 5e). The feedback-region model also outperformed the prediction-region model ($p = 0.047$), indicating that feedback-related neural information was more informative than prediction-related neural information for explaining individual differences in later generalization.

This advantage could not be attributed to differences in voxel count between the mechanism-specific brain masks. In the update-RSA analysis, the feedback mask initially contained fewer voxels than the prediction mask. To rule out a simple volumetric explanation, we conducted a control analysis in which the statistical threshold for the feedback mechanism was adjusted to yield a voxel count closely matched to that of the prediction mask (Supplementary Fig. S6a). Using this voxel-matched feedback mask, we repeated the full individualized modeling procedure. The feedback-aligned model continued to significantly predict Day 7 generalization performance (Spearman $r = 0.41$, $p = 2.37 \times 10^{-5}$; Supplementary Fig. S6d), and in bootstrap comparisons it remained significantly superior to the behavioral-only baseline ($p = 0.0031$; Supplementary Fig. S6e). These results show that the greater contribution of feedback-related neural signals cannot be explained by mask size; rather, it reflects mechanism-specific neural information relevant to inter-learner variability.

Together, these findings show that individualized neurocomputational models can identify the mechanisms driving individual differences in language learning. By retraining participant-specific language models with different neural masks and testing their ability to predict later generalization, we found that feedback-related neural signals were more informative than prediction-related signals for explaining why learners differed in their success in generalization. These results extend the group-level analyses by showing that although prediction and feedback both contribute to language learning, they do so in different ways across levels of analysis. Prediction more strongly supports the emergence of shared representational structure, whereas feedback-related neural information more strongly accounts for variability across individuals in later outcomes.

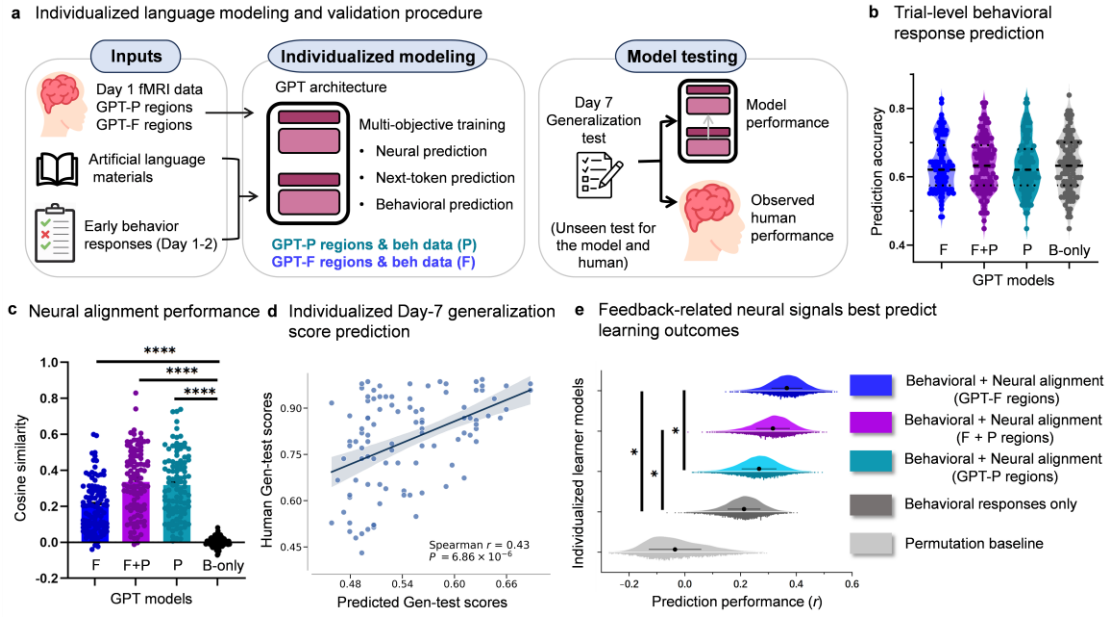


Fig. 5. Individualized models predict later generalization from early behavioral and neural data. **a**, Schematic of the individualized modeling framework. Participant-specific models were trained on artificial-language materials, early behavioral responses and Day 1 fMRI-derived neural targets from prediction-related regions, feedback-related regions or both. **b**, Trial-level behavioral response prediction across individualized models. **c**, Cosine similarity between model representations and participant-specific neural representations. Models trained with neural objectives showed greater neural alignment than the behavioral-only model. ****, $p < 0.0001$. **d**, Prediction of Day 7 generalization scores by the best-performing individualized model. **e**, Bootstrap distributions of prediction performance for generalization. Models using feedback-related neural regions or the combined neural mask outperformed the behavioral-only model, whereas the model using prediction-related neural regions did not. The feedback-region model also outperformed the prediction-region model. *, $p < 0.05$.

Discussion

Using a seven-day artificial-language learning paradigm with fMRI during early learning, and matched transformer models trained with different objectives, we identified a dissociation between the learning signals that best explained the neural structure shared across learners and those that best explained variation in outcomes among learners. At the group level, representations optimized for a prediction objective accounted for the largest portion of unique explainable neural variance, even though participants performed a feedback-based learning task. Throughout training, both prediction-based and feedback-based representations showed a bottom-up shift from sensory regions toward higher-order language and associative networks, with a stronger shift effect for the prediction model than for feedback models. At the individual level, early feedback-related neural responses improved the prediction of Day 7 generalization beyond that of the prediction model and the behavior-only model. Overall, these findings support a multi-signal account of adult language learning, in which prediction primarily shapes the neural representational structure shared across learners, whereas feedback signals more strongly explain why learners later diverge in learning performance.

The group-level findings are most informative when interpreted in light of the learning objectives implemented here. Here, next-token prediction operationalized a prediction-based learning signal, whereas a grammaticality judgment task with trial-by-trial corrective feedback operationalized a feedback-based learning signal. These objectives should not be viewed as complete implementations of predictive-coding theory or reinforcement learning in an algorithmic sense. However, within this operationalization, our results are consistent with prediction-related computations providing a representational scaffold during even early learning. Prediction has been connected to both language learning and processing, with evidence that learners use distributional regularities to anticipate upcoming linguistic structures and that hierarchical predictive signals can be detected across the cortical language system (Caucheteux et al., 2023; Heilbron et al., 2022; Pelucchi et al., 2009; Saffran, 2002; Saffran et al., 1996). At the same time, prediction has also been argued to arise as a byproduct rather than a driver of learning (Chang et al., 2006, 2013). Our results do not fully settle this broader debate, but they do provide an important constraint. Even in a task designed around explicit correctness feedback, representations optimized for prediction showed the strongest model-brain alignment and explained the largest proportion of unique explainable variance. This pattern is more consistent with the view that prediction-related signals are actively involved during learning than with the view that they are merely epiphenomenal.

This interpretation is also consistent with recent work showing that predictive structure is a major source of alignment between deep language models and human brain activity during language processing (Caucheteux et al., 2022, 2023; Goldstein et al., 2022). In that sense, our findings extend the relevance of prediction beyond mature language comprehension to the formation of new linguistic knowledge. Importantly,

the current findings do not imply that prediction alone suffices for effective language learning. Instead, they indicate that prediction-related processes significantly influence the shared neural representational structure across learners.

The complementary role of feedback-related signals is equally important. Adult language learning is often supported by explicit correctness information and outcome-based updating has long been linked to cortico-striatal systems involved in sequential learning, category learning and aspects of grammar acquisition (Feng, Gan, et al., 2021; Feng, Ou, et al., 2021; Krishnan et al., 2016; Schultz et al., 1997). In our data, however, feedback-related representations did not dominate the neural geometry shared across the learners. Instead, they were most informative for individualized prediction. Individualized neural-enhanced language models that incorporated feedback-aligned neural targets predicted Day 7 generalization better than early behavior alone and better than models using prediction-aligned brain regions. This pattern suggests that feedback may be especially informative about how learners use information about errors and correctness to tune their internal representations or decision policies over time. Put differently, prediction-related signals appear to explain more of what learners share, whereas feedback-related signals explain more of what differentiates them.

The stage-specific representational findings suggest that prediction- and feedback-based signals are embedded in a common hierarchical learning trajectory. Across training, model-brain alignment for both objectives shifted from lower-order sensory cortices towards language and transmodal association systems. Early aligned representations were concentrated in sensory regions, consistent with relatively stimulus-bound encoding (Brincat et al., 2018) at the beginning of learning. Later aligned representations became more prominent in language-related and higher-order associative regions, consistent with the emergence of more abstract and integrated representational geometry that can support grammatical structure and generalization (Courellis et al., 2024; Johnston & Fusi, 2023). This interpretation fits prior evidence that language comprehension is organized along a cortical hierarchy, from acoustic and perceptual features to lexical, compositional and contextual representations (Caucheteux & King, 2022; De Heer et al., 2017), and that language-model alignment with the brain is strongest when models capture hierarchical and contextual computations in higher-order language and association networks (Caucheteux et al., 2023; Mischler et al., 2024; Tang et al., 2023). Our results extend this work by showing that a similar sensory-to-association representational progression can be observed during learning itself, not only during mature language comprehension.

Individualized modeling highlights the role of feedback-related neural responses. Early behavioral data poorly predicted later generalization, whereas models using early neural data performed better, especially with feedback-related alignment (outcome prediction improved from $r = 0.27$ to $r = 0.43$). This clarifies the prediction-feedback dissociation, aligning with prior research showing that model-brain alignment varies and that better alignment aids understanding of human language processing (Caucheteux et al., 2022; Schrimpf et al., 2021). Here, we extend that logic to

longitudinal learning outcomes. By combining early behavioral responses with participant-specific neural targets compressed into the model's representational space, we were able to predict later generalization more accurately than early behavior alone. This improvement highlights the importance of neural information in forecasting learning trajectories that develop over days, aligning with previous work demonstrating that early distributed neural patterns contain information about individual language-learning profiles and ultimate achievement (Feng, Ou, et al., 2021; Morgan-Short, 2020; Song et al., 2025).

The prediction-feedback distinction is also psychologically plausible. Prediction-related signals may help learners build structured internal representations that are broadly useful across learning stages and the cohort. Feedback-related signals, by contrast, may be especially sensitive to how individuals use errors, correctness information and performance monitoring to adjust behavior, especially in later training. That kind of adjustment may be the key point at which individual differences are likely to emerge. Some learners may exploit feedback efficiently and convert it into robust generalization, whereas others may rely on it less effectively or in a more item-bound way. Based on this interpretation, feedback carries especially valuable information about individual-level learning efficiency. This is why the contrast should not be framed as one mechanism being globally more important than the other. A better conclusion is that the two signals make distinct contributions to different explanatory targets: one to the neural representations that learners largely share, and the other to the behavioral divergence that emerges between them.

More broadly, these findings clarify the role of model-brain alignment in cognitive neuroscience. While much previous work has used alignment to determine whether artificial models resemble human neural responses during language processing, that question remains largely descriptive. In our context, alignment takes on a more mechanistic role (Kriegeskorte, 2008; Kriegeskorte & Kievit, 2013). By keeping architecture constant and varying the learning objective, we examine how differences in model-brain alignment can shed light on which types of learning pressures best account for various aspects of human learning. Thus, the present study explores not only whether a model is brain-like but also whether model-brain alignment can contribute to understanding how new linguistic knowledge is learned and why learners often differ in their later attainment.

Several limitations qualify these conclusions. First, Brocanto2 offers strong experimental control over input, training history, and grammatical structure, but it is still an artificial language with a limited lexicon and a simplified learning environment (Morgan-Short et al., 2010, 2012). The extent to which the present dissociation generalizes to naturalistic language learning remains open. Second, our comparison is between matched learning objectives, not between exhaustive implementations of full theoretical frameworks. The findings therefore support a dissociation between prediction-based and feedback-based learning signals as operationalized here, rather than definitively resolving predictive-coding theory or reinforcement learning in their

broadest forms. Third, neural data were collected during early learning, whereas learning unfolded over seven days. Repeated neuroimaging across the full training period would provide a stronger test of how the balance between prediction-related and feedback-related signals changes over time. Finally, although humans and models were matched on materials and overall task structure, they were not matched on all aspects of exposure and learning dynamics, such as the number of update steps. Future work should therefore test whether the same pattern holds with richer linguistic input, longer training windows and more naturalistic tasks.

In summary, this study reveals a dissociation between the learning signals that structure shared neural representations and those that explain individual differences in adult language learning. Representations optimized for prediction best accounted for the neural geometry common across learners during early acquisition, whereas feedback-related neural signals more strongly predicted variability in later generalization. These findings argue against a single-mechanism account of adult language learning and instead support a multi-signal framework in which prediction and feedback make complementary and unique contributions at different levels of analysis. More broadly, this work shows that model-brain alignment during learning can move beyond descriptive comparison, providing a principled approach for testing how candidate learning mechanisms shape the acquisition of new linguistic knowledge and why learners differ in their outcomes.

Methods

Participants

One hundred and six healthy adults from communities around South China Normal University (mean age = 21.42 years; 73 females, 33 males) participated. All were native Mandarin speakers with formal training in English as a second language. None had formally studied any Romance language or been exposed to such languages for more than three weeks. Four participants started the experiment, but later withdrew. All reported normal hearing and no history of neurological disorders. The study was approved by the Joint Chinese University of Hong Kong (CUHK) - New Territories East Cluster (NTEC) Clinical Research Ethics Committee (CREC) and the Research Ethics Committee at National Taiwan University. Informed consent was obtained from each participant. Monetary compensation was provided after completion.

Materials

Brocanto2 is a miniature artificial language designed to preserve key properties of natural language while allowing tight experimental control. It is fully productive, can be spoken and comprehended, and embeds lexical and grammatical structures in a meaningful board-game context (Fig. 2a), enabling learners to acquire form-meaning mappings in a constrained yet communicative setting. Prior neuroimaging studies using Brocanto and Brocanto2 have shown natural-language-like neural responses during processing, supporting its use as a controlled model of adult language learning (Friederici et al., 2002; Morgan-Short et al., 2010, 2012; Opitz & Friederici, 2003).

The Brocanto2 lexicon comprised 13 lexical items, including four nouns (pleck, neep, blom, vode), two adjectives with gender-marked forms (troise/troiso, neime/neimo), one gender-marked article (li/lu), four verbs (klin, nim, yab, praz) and two adverbs (noyka, zayma). Nouns referred to game tokens and carried formal grammatical gender; token shape was encoded by adjectives, actions by verbs, and movement direction by adverbs. Articles and adjectives were post-nominal and agreed in gender with the noun, sentences followed a fixed subject-object-verb (SOV) order, and adverbs, when present, immediately followed the verb. To minimize prosodic cues, all words were recorded in isolation and then presented sequentially. Here, the materials were organized into three construction types: noun phrases (NPs), simple subject-verb phrases (SVP), and SOV sentences.

Procedure

Participants completed seven consecutive days of Brocanto2 training. They also conducted one auditory and one speech category learning task for other research purposes. Neuroimaging data were acquired on Day 1, and behavioural training continued across the remaining sessions. Each daily session comprised a vocabulary-exposure phase followed by three grammar-learning tasks targeting NP, SVP, and SOV constructions. On Day 7, participants additionally completed feedback-free generalization tests on novel items. Training and test procedures were implemented in

E-Prime (Psychology Software Tools, Inc.) and were adapted from the Brocanto2 learning paradigm used in earlier work (Feng, Ou, et al., 2021).

At the start of each session, participants completed a passive vocabulary-exposure phase in which each Brocanto2 lexical item was presented auditorily together with its corresponding visual referent. Grammar learning then proceeded as a grammaticality-judgement task with trial-by-trial corrective feedback. No explicit grammatical rules were provided at any point. Instead, participants heard each phrase or sentence while viewing the corresponding game token or movement and judged its acceptability based on immediate intuition. On-screen feedback then indicated whether the response was correct and served as the learning signal for subsequent trials.

In grammar learning, trials were organized into six blocks per session, with two blocks for each type of construction, presented in a fixed order from simpler to more complex structures (i.e., from NP to SVP, then to SOV). Each block included 48 items, half grammatical and half ungrammatical, presented in random order. Ungrammatical items were generated by adding a single violation to an otherwise correct phrase or sentence. Consistent with Brocanto2 morphosyntax, violations targeted gender agreement within NP and word-order rules in SVP and SOV constructions. During learning, the visual display always matched the intended meaning of the auditory item, so the task required grammaticality judgments rather than sentence-picture verification.

Each grammar-learning trial began with a 100-ms fixation cross, followed by a 100-ms blank interval. The auditory stimulus was then presented alongside the corresponding visual display. The duration of the stimulus varied by construction type: 2,400 ms for NPs, 6,300 ms for SVP items, and 7,600 ms for SOV sentences. After the stimulus ended, a question mark prompted the participant to judge grammaticality. Corrective feedback (✓ or ×) was displayed for 1,000 ms after the response. To better distinguish stimulus-related from feedback-related BOLD responses, the interval between response and feedback was jittered from 0 to 4,000 ms, and the inter-trial interval ranged from 2,000 to 6,000 ms.

On Day 7, participants completed a feedback-free grammaticality judgment generalization test that evaluated how well they transferred the learned grammar to new Brocanto2 items. In this test, participants judged 144 previously unseen phrases or sentences, with 72 being grammatical and 72 ungrammatical. During training, each ungrammatical item contained a single violation. The visual displays matched the sentence meaning in all trials, so successful performance depended on identifying grammatical errors rather than semantic mismatches. Participants also took a picture-sentence mapping test on Day 7 to assess whether spoken Brocanto2 sentences accurately described game tokens or movements. The trial structure matched that of the grammaticality judgment task. Although this task was used to evaluate broader generalization, its data were not included in the current analysis.

Image preprocessing

MRI data were preprocessed using fMRIPrep v23.2.1. T1-weighted images were bias-

corrected, skull-stripped, segmented into cerebrospinal fluid, white matter, and gray matter, and normalized to MNI space. For each BOLD run, a reference image was coregistered to the T1-weighted image with boundary-based registration, and head motion was estimated with MCFLIRT. Functional images were resampled to native and MNI space. Nuisance regressors included framewise displacement, DVARS, mean cerebrospinal fluid, white matter, and whole-brain signals, as well as temporal and anatomical CompCor components explaining 50% of the variance. Motion and global signal regressors were expanded with temporal derivatives and quadratic terms, and volumes with framewise displacement > 1.0 mm were flagged as motion outliers. All spatial transformations were applied in a single interpolation step using `antsApplyTransforms` with Lanczos interpolation.

Language model training

We trained three customized GPT-style models using Python’s *transformers* library v4.42.4: GPT-P, GPT-F and GPT-PF. All models shared the same backbone architecture, comprising one embedding layer and two GPT blocks, and differed only in their output heads and training objectives. GPT-style models provide a standard autoregressive architecture for next-token prediction, and the *transformers* library also supports sequence-classification heads for GPT-based models. We used byte-pair encoding for tokenization, together with special tokens such as `<s>` and `</s>`, yielding a vocabulary of 79 tokens.

GPT-P was trained with a next-token prediction objective. GPT-F was trained with a binary classification head to judge whether each input sequence was grammatical or ungrammatical. GPT-F was optimized with cross-entropy loss rather than a standard reinforcement-learning objective (e.g., value learning or policy optimization). Thus, we refer to it as a feedback-based model because its supervision reflects the trial-by-trial correctness feedback in the human task. In the present binary judgment task, explicit correctness feedback served as instructional information (Barto, A. G. & Dietterich, T. G., 2004) that directly specified the desired response rather than functioning only as a scalar reward signal. To include a control model closer to the traditional computational definition of reinforcement learning, we also trained GPT-R using a policy-gradient algorithm (Williams, 1992) (see Fig. S5).

Finally, the GPT-PF combined the next-token prediction and grammaticality classification objectives into one model. The combined loss was defined as $L_{combined} = \alpha L_P + L_F$. To prevent the easier binary classification objective from dominating training, the prediction loss was weighted more strongly when the generation accuracy lagged behind classification performance. Consistent with this rationale, we used $\alpha = 79/2$ when the generation accuracy was lower than the classification accuracy, and $\alpha = 2$ otherwise. This weighting reflects the difference in baseline task difficulty between 79-way next-token prediction and binary grammaticality classification, and also compensates for the smaller amount of grammatical training data available for the prediction objective.

To assess model training performance, we asked GPT-P to generate outputs for all

1,092 items in the exhaustively constructed Brocanto2 corpus. We assessed the model’s generalization ability by measuring the percentage of generated sentences that were both novel and grammatical (i.e., “Gen accuracy”). GPT-F was evaluated by grammaticality-classification accuracy on a held-out 20% validation set, and GPT-PF was evaluated on both metrics.

All three models were trained on datasets intended to approximate the overall scale of human exposure, while differing in the composition of the training input according to their learning objectives. The full training corpus contained grammatical and ungrammatical Brocanto2 sentences in a 1:1 ratio, the same as the human participants encountered during learning. GPT-P was trained only on grammatical sentences, consistent with its generative objective, whereas GPT-F and GPT-PF were trained on 80% of the full corpus, including both grammatical and ungrammatical items. Each model was trained for 200 epochs with a learning rate of 5×10^{-4} , and the checkpoint with the best validation performance was used for subsequent analyses. Training was conducted on a single NVIDIA RTX A6000 GPU.

As an additional control, we trained a prediction model on a corpus comprising the 144 grammatical Brocanto2 items and minimally corrected versions of the 144 ungrammatical items, thereby matching the number of training items in GPT-F. All inputs to this control model were therefore grammatical, as in GPT-P, but the corpus size increased from 144 to 288 items. This control model showed a model training trajectory broadly similar to that of the main GPT-P model and reached a peak generation accuracy of 0.94 (Supplementary Fig. S4).

Model behavior and changes in internal representations

To test whether updates in model representations predicted changes in model behavior during training, we analyzed checkpoints prior to the best-performing checkpoint for each model. For each model, we repeated a resampling procedure 1,000 times. In each repetition, 12 checkpoints were randomly sampled, yielding 66 checkpoint pairs. For each checkpoint, all 288 Brocanto2 items were passed through the model, and the final-token hidden-state vector was extracted from each layer, including the embedding and the two GPT layers.

For each checkpoint pair, we quantified representational updating by subtracting the item-wise hidden-state vectors at the earlier checkpoint from those at the later checkpoint. This produced a representational-change vector for each item. We then computed an item-by-item representation-update RDM using Euclidean distance between these change vectors.

Model behavior was quantified on the same 288 items and using the same checkpoint pairs. For GPT-P, behavioral updating was defined as the change in item-level sentence-generation probability, computed as the mean sentence log probability. For GPT-F, behavioral updating was defined as the change in grammaticality-judgment output (i.e., grammatical or ungrammatical). For GPT-PF, behavioral updating was computed separately for the generation branch and the grammaticality-judgment branch,

and also for a combined measure obtained by averaging the z-scored changes from the two branches. For each behavioral measure, we constructed an item-by-item behavior-update RDM using Euclidean distance.

Finally, for each checkpoint pair and layer, we correlated the representational-update RDM with the corresponding behavior-update RDM using Spearman correlation. Within each repetition, correlations were averaged across the 66 checkpoint pairs to obtain one summary value per layer and behavioral measure. The resulting 1,000 resampled values were used to estimate the stability of the representation-behavior coupling and to compare this coupling across layers and models.

Model-brain alignment with representational similarity analysis (RSA)

We used RSA to compare the geometry of the model and neural representations during language learning. RSA provides a common framework for comparing representational structures across computational models and brain activity, and Euclidean distance is widely used in searchlight-based multivoxel analyses when both spatial patterns and response magnitudes may carry information. For each Brocanto2 phrase or sentence, the full token sequence was passed through each trained model, and the hidden state at the final token was extracted from each model layer (the embedding layer plus two GPT hidden layers) as the sequence representation. Layer-specific model representational dissimilarity matrices (RDMs) were then built across stimuli using cosine distance. To analyze training-related representational changes, we selected four checkpoints along each model's performance trajectory, yielding three consecutive training stages. For GPT-P, these checkpoints corresponded to generation accuracy of 0.33, 0.52, 0.71, and 0.91; for GPT-F, they matched classification accuracies of 0.51, 0.67, 0.83, and 1.00. At each stage, we calculated checkpoint-to-checkpoint difference vectors in hidden-state space and built representational-update RDMs using Euclidean distance, which measures the extent of representational change across training.

For the neural data, stimulus-specific t maps were estimated with Nilearn (v0.10.3), and a neural RDM was computed for each participant. Whole-brain RSA was implemented with a searchlight procedure in native space using a 10-mm spherical region centered on each voxel; within each searchlight, Euclidean distance quantified dissimilarity among local multivoxel activity patterns, and the resulting maps were transformed to standard space. We then performed two multiple-regression RSA analyses. In the cross-layer analysis, each neural RDM was regressed onto the two layer-specific RDMs from the same model. In the representational-update analysis, each neural RDM was regressed onto the three stage-specific update RDMs from a single layer. In both analyses, we included three control RDMs: an untrained GPT exposed to the same inputs, to account for representational structure attributable to architecture and stimulus exposure alone; a stimulus-length RDM, to account for variance related to sequence length and associated processing demands; and a block-identity RDM, to account for variance attributable to the experimental block structure. These controls were included to isolate model-brain alignment specific to the learned

representational geometry rather than to generic architectural or design-related similarities. Group-level significance was assessed with threshold-free cluster enhancement (TFCE) and permutation testing (10,000 permutations; family-wise error-corrected $p < 0.05$) (Nichols & Holmes, 2002; Smith & Nichols, 2009).

To characterize alignment dynamics across brain network systems, we additionally performed network-level RSA as part of the representational update analysis. The brain was parcellated into 12 functional networks using the Ji et al. atlas (2019). Neural RDMs were computed within each network using Euclidean distance, and alignment with the three stage-specific update RDMs was assessed with partial Pearson correlations. This analysis estimated the unique contribution of each training stage while controlling for the other two stages.

Individualized neurocomputational modeling with neural, behavioral, and language data

Unlike earlier GPT models, which were optimized for corpus-level performance, this individualized model was designed to capture participant-specific behavioral and neural structures to predict subsequent generalization outcomes and to examine predictive or feedback-based neural signals that explain future learning outcomes.

To incorporate neural signals into the models, we trained a self-supervised autoencoder to compress each participant's high-dimensional brain data into a lower-dimensional representation matching the dimensionality of GPT's hidden states. This compressed neural representation was then used as an additional training target during model retraining. Although linear mapping approaches are commonly used to relate brain activity to computational models, they are limited in capturing nonlinear structure in large-scale neural data. By contrast, the autoencoder-based approach preserved the underlying nonlinear geometry of neural representations and, crucially, enabled joint optimization of multiple objectives within a single framework: predicting behavior, predicting the next word and approximating neural activity (Han et al., 2019; Huang et al., 2018). ROIs were defined using voxel-wise masks derived from significantly activated regions in the third stage of the representational-update analysis. Separate personalized models were then retrained using prediction-related masks, feedback-related masks or their combination, allowing us to test which type of neural information best accounted for inter-learner variability (Fig. 5a).

For each participant, we trained a participant-specific GPT model with three objectives: predicting that participant's grammaticality judgments during early learning (Days 1-2), fitting early neural responses from selected brain regions (i.e., brain areas sensitive to predictive learning vs. feedback learning signals), and predicting the next token in Brocanto2 sequences. Because the neural data in standard space were far higher-dimensional than the model hidden state (271,633 voxels versus 768 units), we first compressed the neural data with an autoencoder (Han et al., 2019; Huang et al., 2018). The autoencoder comprised three fully connected layers, with a 768-dimensional bottleneck matched to the GPT hidden size. For each participant, it was trained on the

other 101 participants in a leave-one-subject-out procedure and then applied to the held-out participant to generate low-dimensional neural targets. This approach ensures dimensionality transformation while preserving as much information as possible from the original data.

The individualized GPT was optimized with a weighted multitask objective, $L_{\text{combined}} = \alpha L_{\text{behavior}} + \beta L_{\text{neural}} + \gamma L_{\text{predict}}$, where L_{behavior} was the cross-entropy loss for participant-specific judgement prediction, L_{neural} was the mean-squared error between the model’s final-layer hidden states and the compressed neural data, and L_{predict} was the cross-entropy loss for next-token prediction. This weighted-sum formulation is standard in multitask learning when objectives differ in type and scale, but the exact coefficients are heuristic balancing hyperparameters rather than quantities with a unique theoretical optimum (Ruder, 2017). We set $\gamma = 1$ as the reference weight for the language-model objective. We set $\alpha = \ln(79/2)$ to upweight the behavioral term relative to next-token prediction, linking the weight to the difference in baseline cross-entropy between a 79-way token-prediction problem and a binary judgment problem. We set $\beta = 4/768$ as a small scaling factor to keep the neural regression term numerically comparable to the two cross-entropy losses. Behavioral supervision used each participant’s observed responses on Days 1 and 2; to reduce day-specific overfitting, training alternated between Day-1 and Day-2 response labels every 30 epochs.

For each participant, Day-1 and Day-2 data were split into 70% for training and 30% for held-out validation. Models were trained for 100 epochs, and the checkpoint with the highest held-out behavioral accuracy was retained. The trained models saved in this checkpoint were then used to predict grammaticality judgments in the Day-7 generalization task, testing whether early behavioral and neural signals forecast later responses to novel items for each individual learner. A significant correlation between model-predicted and observed generalization performance was taken as evidence of successful individualized modeling and suggested that information from the corresponding brain system contributed to predicting individual differences in learning outcomes.

Data availability

The data will be available upon request from the corresponding author.

Code availability

The preprocessed code for replicating the main analyses of this paper are available at:

https://github.com/thedarkkinght/GPTs_learn_Brocanto2

Author contributions: Shu.Y. and G.F. designed research; Shu.Y., X. J., and S.W. performed research; Shu.Y. and Shao.Y. analyzed data; Shu.Y. and G.F. wrote the first draft of the paper; Shu.Y., Shao.Y., and G.F. edited the paper; Shu.Y. and G.F. wrote the paper.

Supplementary Materials

Supplementary Figures

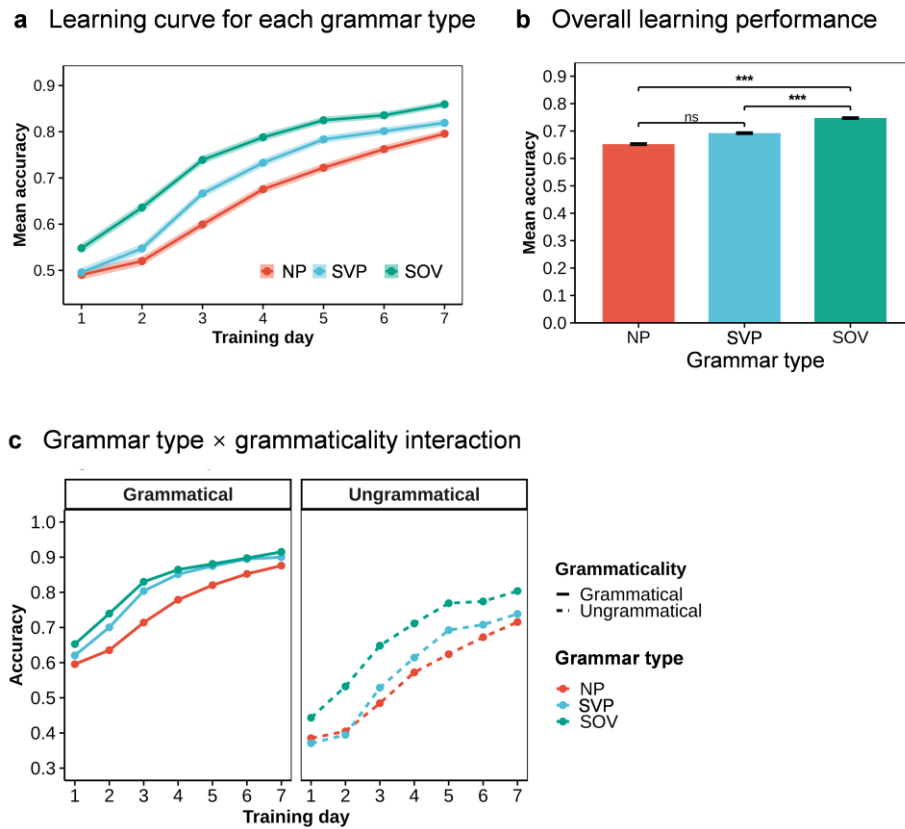


Fig. S1. Behavioral learning across training days. Trial-level accuracy was analyzed with a mixed-effects logistic regression including random intercepts for participants and items. **a**, Accuracy improved across the 7-day training period for all construction types, with the highest performance for SOV, followed by SVP and NP. **b**, Main effect of construction type. Bonferroni-corrected pairwise comparisons showed that SOV accuracy was higher than SVP ($\beta = 0.52$, $p < 0.001$, OR = 1.69) and NP ($\beta = 0.55$, $p < 0.001$, OR = 1.74), whereas SVP and NP did not differ. **c**, Construction type × grammaticality interaction. Accuracy was higher for grammatical than ungrammatical items overall ($\beta = 1.09$, $p < 0.001$, OR = 2.96), and this grammaticality effect differed across construction types ($\chi^2(2) = 6.18$, $p = 0.045$). Accuracy also increased reliably across days ($\beta = 0.29$, $p < 0.001$), with faster improvement for SVP and SOV than NP (SVP × Day: $\beta = 0.042$, $p < 0.001$; SOV × Day: $\beta = 0.031$, $p < 0.001$).

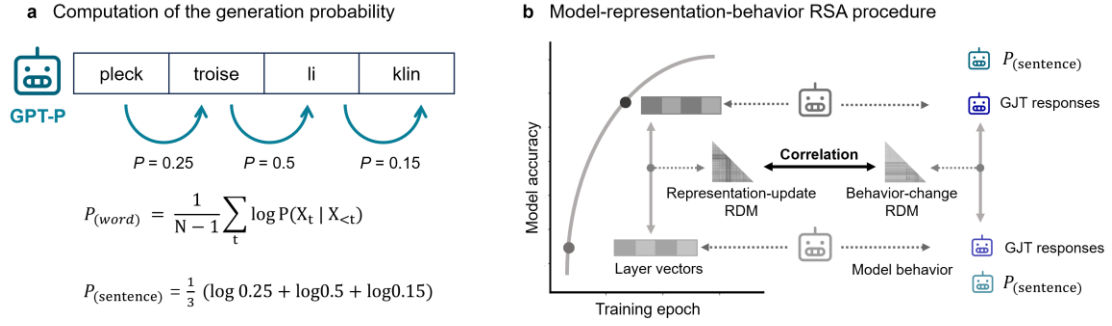
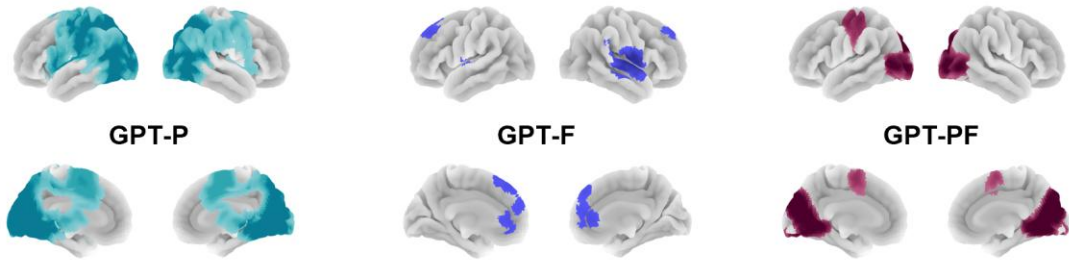
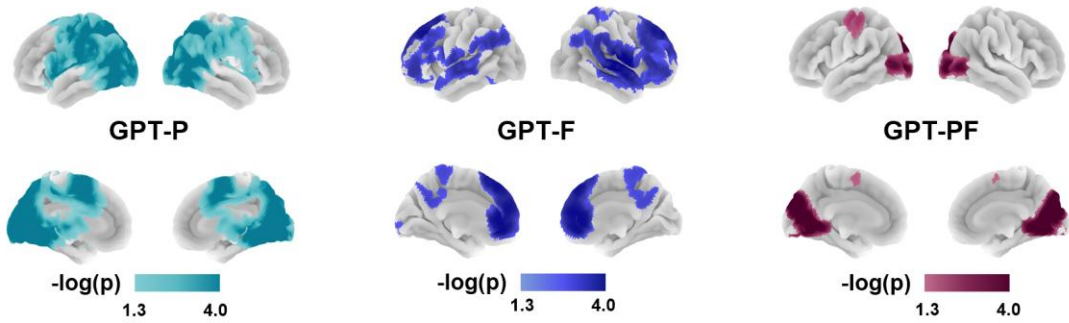


Fig. S2. Similarity analysis procedure to examine the relationship between model representational updates and behavioral changes. **a**, Item-level behavioral scores for GPT-P were computed from the mean sentence log probability, obtained by averaging token-wise conditional log probabilities across each Brocanto2 sequence. This score indexed the model's generative behaviors for each item. **b**, Model-representation-behavior RSA procedure. For each pair of training checkpoints, representational updating was computed as the item-wise difference between layer vectors for each model layer. Behavioral updating was computed using the same checkpoint pair from item-level model outputs: mean sentence log probability for GPT-P, grammaticality judgment responses for GPT-F, and the corresponding generation, classification, and combined scores for GPT-PF. Representational-update and behavior-update RDMs were constructed across the 288 Brocanto2 items using Euclidean distance, and their correspondence was assessed using Spearman's rank correlation.

a Model-brain alignment (Layer 1)



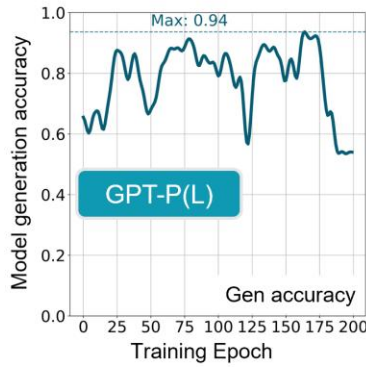
b Model-brain alignment (Layer 2)



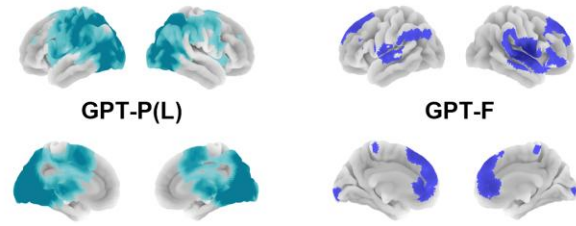
(FWE-corrected $p < 0.05$)

Fig. S3. Layer-wise model-brain representational alignment. **a,b**, Whole-brain searchlight RSA results for the first (a) and second (b) hidden layers of GPT-P, GPT-F and GPT-PF. Model RDMs were fitted to fMRI-derived neural RDMs while controlling for an untrained GPT baseline, stimulus length and block identity. GPT-P showed broad alignment across both layers. GPT-F showed weaker, more spatially restricted alignment in layer 1 but stronger alignment in layer 2. GPT-PF showed significant alignment in both layers, with a more focal spatial pattern in sensorimotor regions. Brain maps were thresholded using TFCE with 10,000 permutations and family-wise-error correction at $p < 0.05$. Color bars indicate $-\log(p)$.

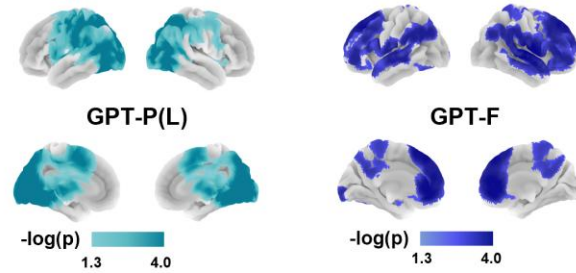
a GPT-P(L) training trajectory



b Model-brain alignment (Layer 1)



c Model-brain alignment (Layer 2)



(FWE-corrected $p < 0.05$)

Fig. S4. Control prediction model trained on an expanded grammatical corpus. **a**, training trajectory of GPT-P(L), a prediction model trained on the original 144 grammatical Brocanto2 items plus 144 minimally-corrected grammatical versions of ungrammatical items. GPT-P(L) showed a model training trajectory similar to the main GPT-P model and reached a peak generation accuracy of 0.94. **b**, **c**, Whole-brain model-brain alignment results for layer 1 (**b**) and layer 2 (**c**). GPT-P(L) was compared with the same GPT-F models used in the main analysis. Consistent with the main prediction-model findings, GPT-P(L) showed robust and spatially extensive alignment with human neural representations across layers. Brain maps were thresholded using TFCE with 10,000 permutations and family-wise-error correction at $p < 0.05$. Color bars indicate $-\log(p)$.

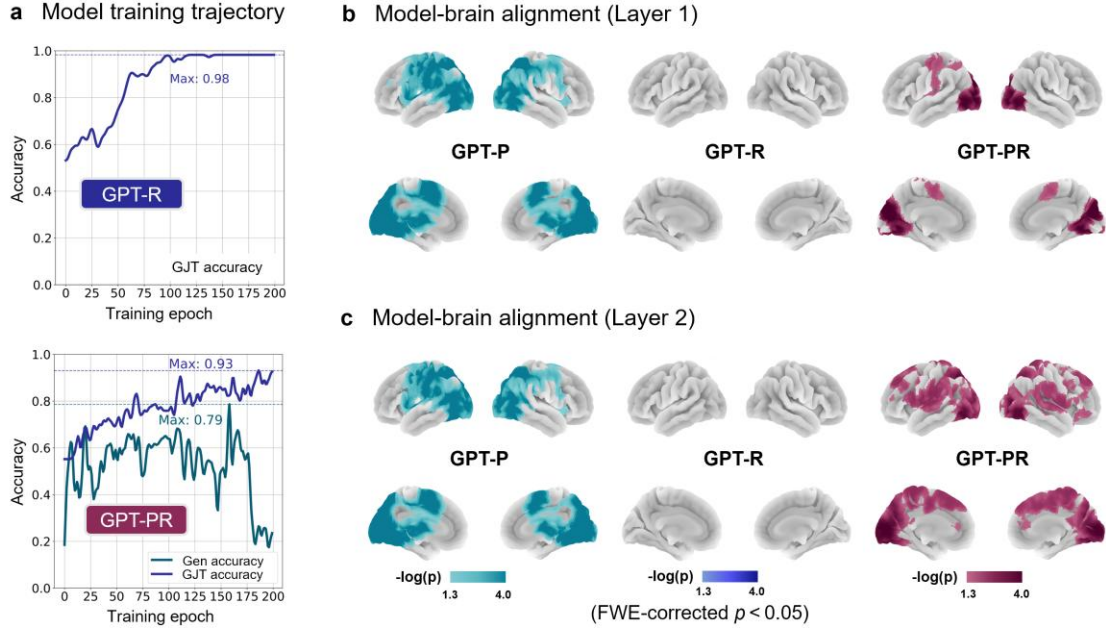


Fig. S5. Policy-gradient control models and model-brain alignment. **a**, Validation performance across training epochs for the policy-gradient grammaticality model, GPT-R, and the combined prediction-plus-policy-gradient model, GPT-PR. GPT-R was evaluated by grammaticality-classification accuracy and reached a maximum accuracy of 0.98. GPT-PR was evaluated on both grammaticality classification accuracy and generation accuracy, achieving a maximum classification accuracy of 0.93 and a maximum generation accuracy of 0.79. **b**, **c**, Whole-brain RSA results for the first (**b**) and second (**c**) hidden layers. GPT-P showed significant model-brain alignment in both layers, whereas GPT-R showed no suprathreshold alignment despite high task accuracy. GPT-PR showed significant alignment, particularly in the second layer. Brain maps were thresholded using TFCE with 10,000 permutations and family-wise-error correction at $p < 0.05$. Color bars indicate $-\log(p)$; grey indicates no suprathreshold voxels. These results indicate that policy-gradient task learning alone was insufficient to reproduce human neural representational geometry, whereas alignment emerged when policy-gradient learning was combined with a prediction objective.

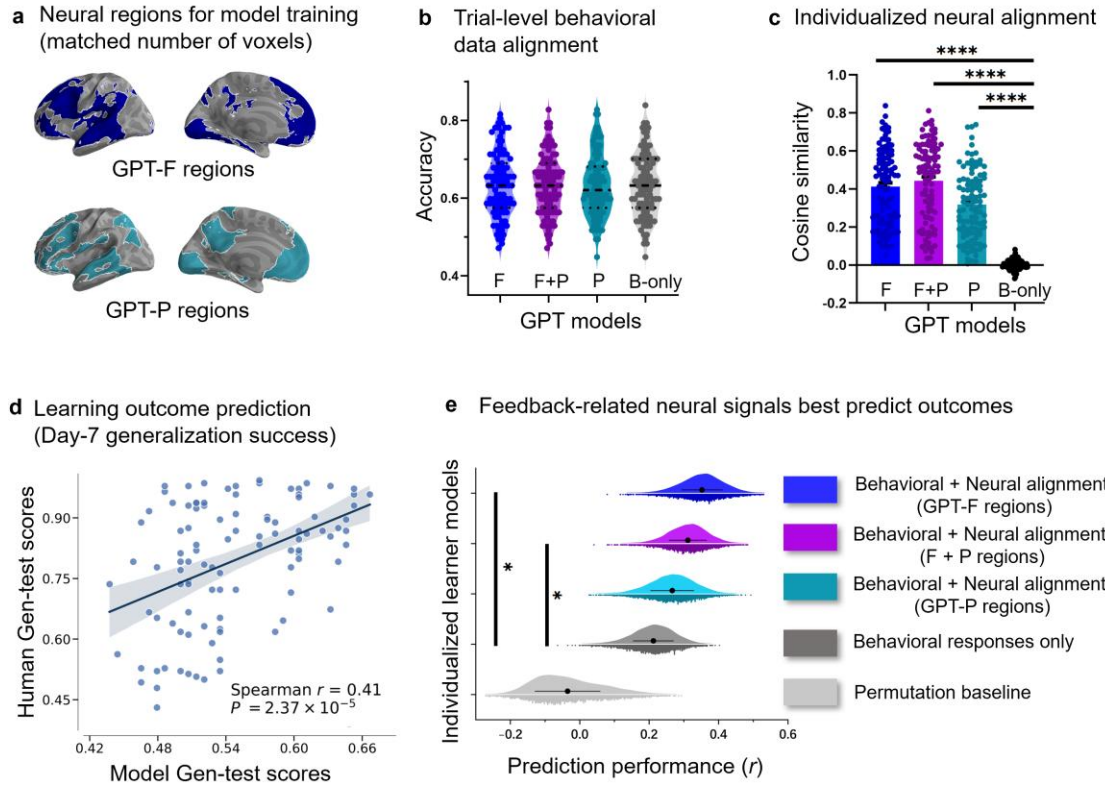


Fig. S6. Voxel-matched control analysis for individualized model prediction. **a**, Prediction- and feedback-aligned neural masks (derived from the update RSA results) used for individualized model training. The feedback-aligned mask was threshold-adjusted to match the voxel count of the prediction-aligned mask. **b**, Trial-level consistency between individualized model predictions and participants' grammaticality judgments during early training (i.e., days 1 and 2). **c**, Neural alignment of individualized models, measured as cosine similarity between model hidden representations and participant-specific compressed neural targets. Bars show group means; dots indicate participants. **d**, Association between model-predicted and observed Day 7 generalization performance. Each dot represents a participant; the line shows the fitted association. Spearman's r and p values are reported. **e**, Bootstrap distributions of prediction performance across 10,000 resamples. Asterisks indicate models whose median correlation exceeded the 95th percentile of the behavioral-only baseline.

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